



# JURONG JUNIOR COLLEGE

## JC2 Preliminary Examination 2018

CANDIDATE  
NAME

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CLASS

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INDEX  
NUMBER

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### BIOLOGY

**9744/04**

Paper 4 Practical

**14 August 2018**

Candidates answer on the Question Paper

**2 hours 30 minutes**

Additional Materials: As listed in the Confidential Instruction

#### READ THESE INSTRUCTIONS FIRST

Write your class, index number and name on all the work you hand in.  
Give details of the practical shift and laboratory, where appropriate, in the boxes provided.  
Write in dark blue or black pen.  
You may use an HB pencil for any diagrams or graphs.  
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.  
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.  
The number of marks is given in brackets [ ] at the end of each question or part question.

|                   |
|-------------------|
| <b>Shift</b>      |
|                   |
| <b>Laboratory</b> |
|                   |

| For Examiner's Use |  |
|--------------------|--|
| <b>1</b>           |  |
| <b>2</b>           |  |
| <b>3</b>           |  |
| <b>Total</b>       |  |

This document consists of **20** printed pages.

Answer **all** questions.

- 1** Protease enzymes catalyse the hydrolysis of proteins into smaller, more soluble molecules.

In this investigation, the substrate for protease is the protein casein found in milk.

When the enzyme and milk are mixed, the casein is hydrolysed and the mixture will change from cloudy to clear.

You will record this change using the ten-point scale shown.

|                                                                                                 |   |   |   |   |   |   |   |   |   |   |
|-------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|---|
| 10                                                                                              | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| cloudy  clear |   |   |   |   |   |   |   |   |   |   |

You are required to investigate the effect of the enzyme concentration on the hydrolysis of casein.

You are provided with the following:

- 60 cm<sup>3</sup> of milk containing casein, labelled **M**,
- 15 cm<sup>3</sup> of protease solution, labelled **E1**,
- 15 cm<sup>3</sup> of protease solution, labelled **E2**,
- 15 cm<sup>3</sup> of protease solution, labelled **E3**,
- 30 cm<sup>3</sup> of distilled water, labelled **W**.

Solutions **E1**, **E2** and **E3** are different concentrations of protease.

Protease is harmful. You are advised to wear the eye protection and gloves provided, and wash off any spillages.

*Proceed as follows.*

### Stage 1

Use the beaker or container provided to make a water-bath with warm water, at a temperature between 35 °C and 40 °C.

Maintain the temperature of the water-bath throughout the procedure.

Read through **stages 2–4** and prepare a table to record your results in **stage 5** before proceeding further.

### Stage 2

Label five boiling tubes, **B1**, **B2**, **B3**, **B4** and **B5**.

Put 5.0 cm<sup>3</sup> of **M** into each of the five boiling tubes.

- (a) Using the ten-point scale, record the initial appearance of the mixtures in the boiling tubes in the table you have prepared in **stage 5**. [1]

### Stage 3

This stage must be completed quickly since the reaction will start as soon as protease is added.

Put 1.0 cm<sup>3</sup> of the protease solutions, **E1**, **E2** and **E3** or distilled water, **W**, into each boiling tube, as shown in Table 1.1.

**Table 1.1**

| <b>B1</b>                       | <b>B2</b>                       | <b>B3</b>                       | <b>B4</b>                      | <b>B5</b>                       |
|---------------------------------|---------------------------------|---------------------------------|--------------------------------|---------------------------------|
| 1.0cm <sup>3</sup> of <b>E1</b> | 1.0cm <sup>3</sup> of <b>E2</b> | 1.0cm <sup>3</sup> of <b>E3</b> | 1.0cm <sup>3</sup> of <b>W</b> | 1.0cm <sup>3</sup> of <b>E3</b> |

Gently mix the solution in each boiling tube, taking care to avoid splashes. Put the boiling tubes into the water-bath and immediately start a stopwatch.

### Stage 4

Every 2 minutes, use the ten-point scale to record the appearance of the mixtures in boiling tubes **B1**, **B2**, **B3**, **B4** and **B5** in the table you have prepared in **stage 5**. After 10 minutes, stop recording.

### Stage 5

- (b) Use this space to draw your table of results. [5]

| Tube/mixture/solution | Appearance of tube/mixture (at given time interval/at min) |       |       |       |       |        |
|-----------------------|------------------------------------------------------------|-------|-------|-------|-------|--------|
|                       | 0 min OR<br>Initial appearance                             | 2 min | 4 min | 6 min | 8 min | 10 min |
| B1                    | 10                                                         | 8     | 8     | 7     | 5     | 5      |
| B2                    | 10                                                         | 6     | 4     | 3     | 2     | 2      |
| B3                    | 10                                                         | 4     | 3     | 2     | 1     | 0      |
| B4                    | 10                                                         | 10    | 10    | 10    | 10    | 10     |
| B5                    | 10                                                         | 4     | 3     | 2     | 1     | 0      |

1. Table heading- tubes + appearance/observation of cloudiness/scale, min reading/degree of cloudiness ; R: observation
2. Columns/rows to show data at every 2 min interval ;
3. All data recorded ;
4. Records using ten-point scale (0-10) ; R: colour
5. Quality of data: scale point decreases over time/plateau for B1- B3 and B5 (2min vs 10min) + control = 10 ;

(c) Explain the purpose of including boiling tubes **B4** and **B5** in the investigation. [2]

**B4: as a control to prove that the change in appearance of the content is due to the activity of an enzyme ;**

**B5: replicate reading/repeat to ensure that results are reproducible/ensure reliability ;**

(d) Identify **one** significant source of error in using the ten-point scale to record the appearance of the mixtures. [1]

- 1. Difficulty in ensuring the same level of cloudiness is obtained at a given index ; *owtte***
- 2. Subjectivity in judgement of cloudiness ;**
- 3. No colour standard used in comparison ;**

(e) Describe **three** modifications to your investigation which would improve confidence in your results. [3]

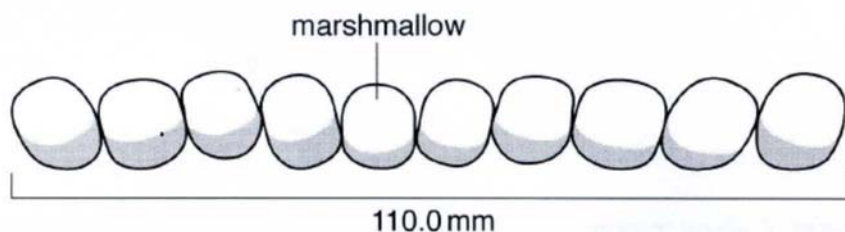
- 1. Use colourimeter ;**
- 2. Repeat each concentration of protease ;**
- 3. Stagger the start time for each concentration ;**
- 4. Use a thermostatically controlled water bath ;**

**Turn over for the remainder of Question 1**

Marshmallows are made using the protein gelatine, which is hydrolysed by a protease enzyme.

Students investigated the activity of protease at different pH values, starting with pH 1.0.

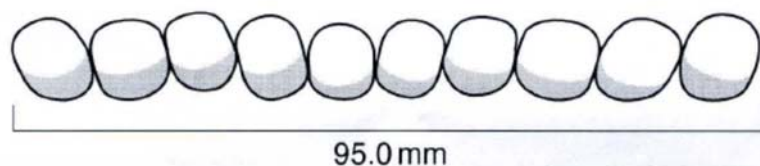
10 marshmallows were arranged in a line and their combined length was measured as shown in Fig. 1.1.



**Fig. 1.1**

The marshmallows were placed in a Petri dish and covered with a solution of protease maintained at pH 1.0.

After 60 minutes the marshmallows were measured again, as shown in Fig. 1.2, and the decrease in their combined length calculated.



**Fig. 1.2**

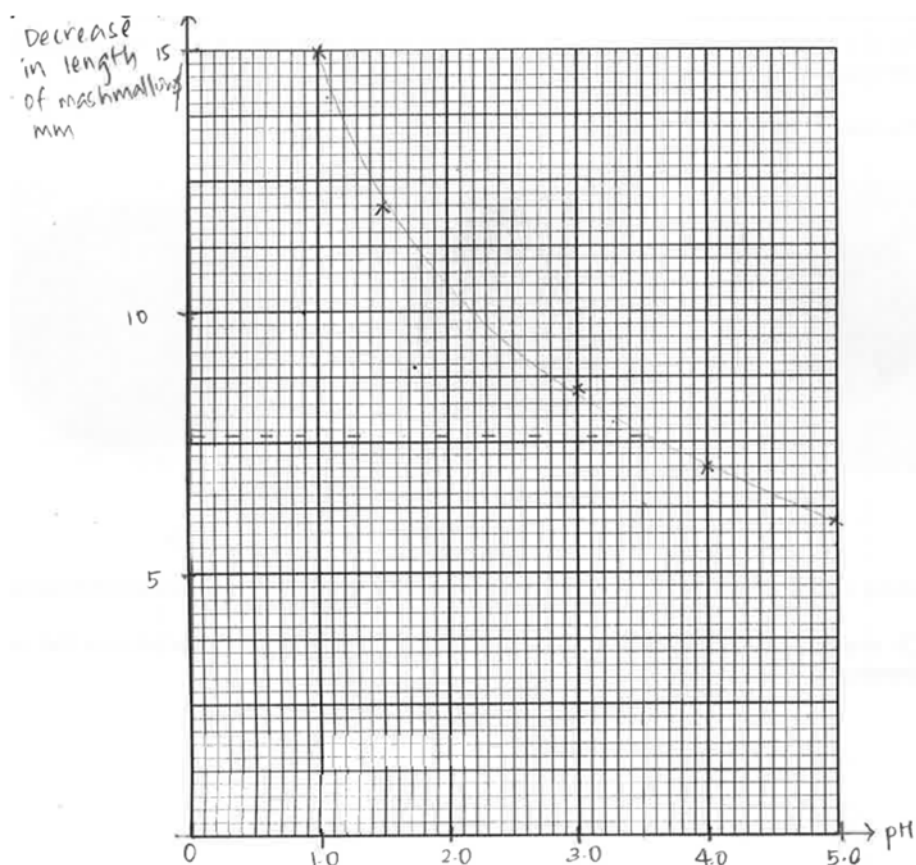
The procedure was repeated with four more sets of marshmallows and solutions of protease maintained at pH 1.5, 3.0, 4.0 and 5.0.

The results of the students' investigation are shown in Table 1.2.

**Table 1.2**

| pH  | decrease in combined length of marshmallows/mm |
|-----|------------------------------------------------|
| 1.0 | 15.0                                           |
| 1.5 | 12.0                                           |
| 3.0 | 8.5                                            |
| 4.0 | 7.0                                            |
| 5.0 | 6.0                                            |

(f) Plot a graph of the data shown in Table 1.2. [4]



1. Criterion 1

Sensible scales must be used. Awkward scales (e.g. 3:10) are not allowed. Scales must be chosen so that the plotted points occupy at least half the graph grid in both x and y directions ;

2. Criterion 2

Correct choice of axis and complete with labels ;  
y axis: decrease in length of marshmallow/mm  
x axis: pH

3. Criterion 3

All data must be plotted. Check all points and put ticks if correct ;

4. Criterion 4

Curve or point-to-point plot drawn ;

(g) (i) Use your graph to estimate the decrease in combined length of marshmallows at pH 3.5.

Show on your graph how you estimated the decrease in length.

- Shows dotted line on graph at pH 3.5 to read off length on y-axis ;
- Correct value + units in mm ;

decrease in length = **7.6mm** [2]

(ii) Explain the reasons for the differences in the decreases in length at low pH (between 1 and 2) and at higher pH (between 3 and 5). [3]

1. Decrease in length was greater at low pH than high pH/steep decrease at low pH vs gradual decrease at high pH ;
2. low pH is optimum for protease where maximum rate of reaction occurs/ highest frequency of successful collisions between the substrate(s) and enzyme/ high rate of formation of E-S complexes or product ;
3. high pH alters the ionic charge of/ionizes (of the acidic and basic R groups on the) amino acids at the active site /enzyme ;
4. The ionic bonds and hydrogen bonds that help to maintain the specific shape of the active site are broken/ loss of the specific 3D conformation of the enzyme's active site/enzyme is denatured ;

[Total: 21]



**2** During this question you will require access to a microscope and slide **B1**.

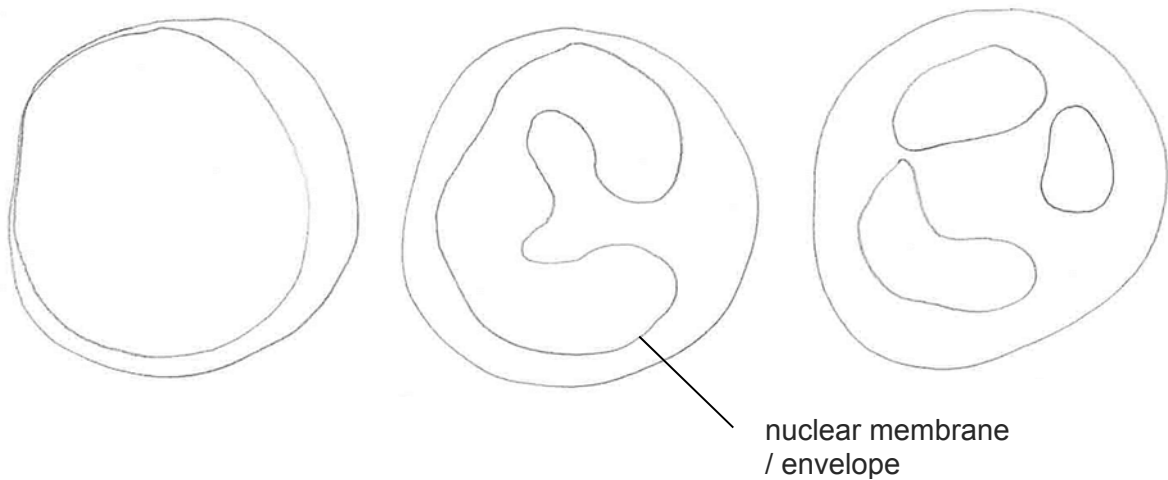
Slide **B1** shows white blood cells and red blood cells. You are not expected to be familiar with these cells.

**(a)** Examine the slide under the high-power objective lens of your microscope and locate the white blood cells.

Make a detailed drawing of **two** white blood cells in the spaces provided.

The drawings should show differences in appearances between each type of white blood cell.

On one of your drawings, use a ruled label line and label to show the position of the nuclear membrane. [5]



1. **Quality of drawing:** clear, sharp, continuous line, with no shading, drawing occupies at least half the space given ; (allow 1 error)
2. **Draws 2 cells ; R: more or less cells**
3. **Correct proportion of nucleus (fills ~50% of whole cell) ;**
4. **(Details of) nucleus different between two types of WBC (lobes nucleus) ;**
5. **Correct label: nuclear membrane/envelope ;**



- (b) (i) Using the objective lens selected in (a), complete Table 2.1 so that in (b)(ii) the **mean** number of red blood cells in the field of view can be calculated.

Count every red blood cell for which the **entire cell** is visible in the field of view. [1]

**Table 2.1**

|                                              |
|----------------------------------------------|
| number of red blood cells in a field of view |
| <b>At least 3 different counts;</b>          |

- (ii) Calculate the mean number of red blood cells per field of view, using your data from Table 2.1.

Show your working.

- Correct display of calculation of mean i.e show working**  
**→ Add up the 3 results in Table 2.1 / 3 ;**
- Correct calculation of mean no. of rbc to nearest whole number ;**  
*expected 1000-5000*

mean number of red blood cells per field of view = ..... [2]

(c) Focus the objective lens that you used for counting red blood cells on the stage micrometer provided.

(i) Measure the radius of the field of view and record this in mm.

1. Answer to 2 d.p.

$r = 0.22$  or  $0.23$  mm (under 40x objective)

radius,  $r$ , of field of view = ..... mm [1]

(ii) Calculate the area of the field of view and record this area in  $\text{mm}^2$ .

Area of circle =  $\pi r^2$ , given  $\pi = 3.142$

1. Correct calculation of area of field of view using value of  $r$  from (c)(i) to 2 or 3 d.p ; A : scientific notation

Area of field of view = .....  $\text{mm}^2$  [1]

(d) Using your results from (b)(ii) and (c)(ii), calculate the mean density of red blood cells per  $\text{mm}^2$ .

1. Correct calculation of mean density of red blood cells to nearest whole no.

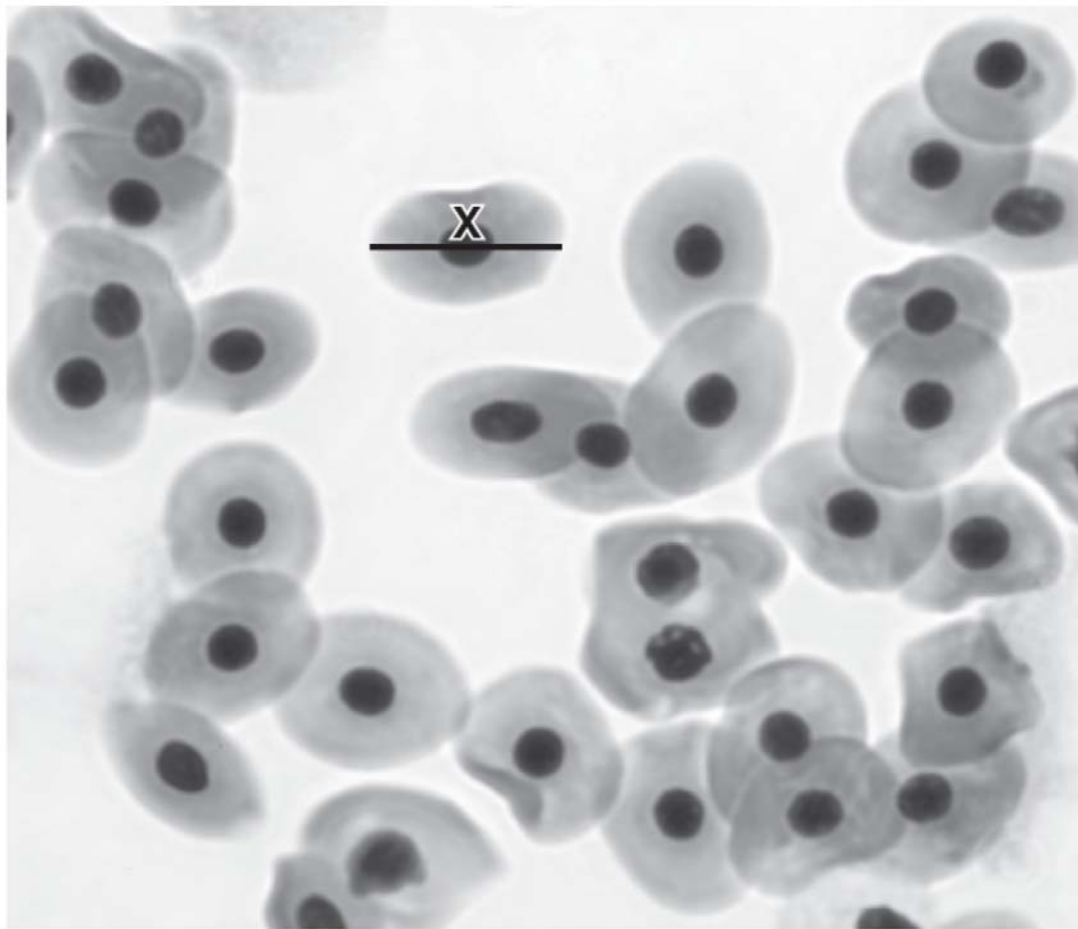
→ (b)(ii) results / (c)(ii) results ;

mean density of red blood cells = .....  $\text{mm}^{-2}$  [1]

**Turn over for the remainder of Question 2**

Fig. 2.2 is a photomicrograph of blood taken from another type of organism.

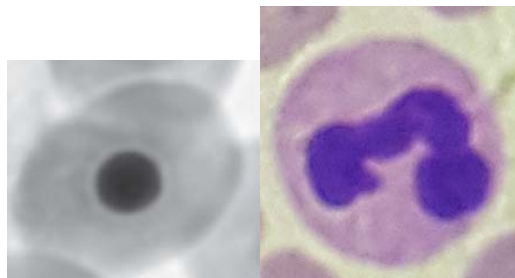
You are not expected to be familiar with these cells.



magnification  $\times 700$

**Fig. 2.2**

- (e) Prepare the space below so that it is suitable for you to compare and contrast the blood cells in Fig. 2.2 and the blood cells drawn in (a). [5]



1. (organise) table + heading for differences (Fig 2.2 and (a)) + statements opposite each other + feature column ;

| Feature                                                                                                                                                                                                                                                                                                                                                                                                             | Fig. 2.2                                                                                                                                                                | (a)                                                                                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>D1. Cell shape</b>                                                                                                                                                                                                                                                                                                                                                                                               | oval /<br>some or few round / circular<br><br>R: “less circular” / “not circular” / “spherical” / “biconcave” / “disc” / “rugby” / “irregular” / “arbitrary” / “random” | circular / irregular / round                                                                                                                                                                                      |
| <b>D2. Cytoplasm</b>                                                                                                                                                                                                                                                                                                                                                                                                | not granular                                                                                                                                                            | granular                                                                                                                                                                                                          |
| <b>D3. Shape of nucleus<br/>OR<br/>Number of nuclei (Difference)</b><br><br><ul style="list-style-type: none"> <li>Award once for either “shape” or “number”</li> <li>BOD for wrong match between “shape” / “number” and respective descriptors but contrasting adjectives must be correctly paired</li> <li>Reject: “Types / variety / appearance / position of nucleus” even with matching descriptors</li> </ul> | oval / round / regular<br><br>one / singular<br><br>R: “same type” / “one variety” / “have 1 type” / “nucleus in the centre”                                            | lobed / irregular /<br><br>more than one (AW) / a few / multinucleated / segmented<br><br>R: “fragmented” / “segregated” / “divided” / “varied” / “have a variety” / “have 2 types” / “nucleus not in the centre” |
| <b>D4. Density of cells<br/>OR<br/>Number of cells<br/>I: “white blood cells”<br/><br/>R: “spacing between cells” / “arrangement” / “clumping”</b>                                                                                                                                                                                                                                                                  | lower / lesser / less dense / sparse<br><br>more / a lot of / many / numerous blood cells<br><br>R: “overlapping” / “clumped”                                           | higher / more / more dense<br><br>less / fewer blood cells<br><br>R: “non-overlapping” / “not clumped”                                                                                                            |
| <b>D5. Size of nucleus<br/>OR<br/>nucleus:cytoplasm ratio</b>                                                                                                                                                                                                                                                                                                                                                       | smaller<br><br>lower / smaller                                                                                                                                          | larger / bigger / greater<br><br>higher                                                                                                                                                                           |
| <b>S1. Presence of nucleus<br/>OR<br/>Number of nuclei (Similarity)</b>                                                                                                                                                                                                                                                                                                                                             | present<br><br>both have one nucleus each [must agree with drawing in (a)]                                                                                              |                                                                                                                                                                                                                   |
| <b>S2. Presence of cytoplasm</b>                                                                                                                                                                                                                                                                                                                                                                                    | present                                                                                                                                                                 |                                                                                                                                                                                                                   |
| <b>S3. Presence of cell membrane</b>                                                                                                                                                                                                                                                                                                                                                                                | present<br>(award once for S3 or S4)                                                                                                                                    |                                                                                                                                                                                                                   |
| <b>S4. Presence of nuclear membrane</b>                                                                                                                                                                                                                                                                                                                                                                             | present<br>(award once for S3 or S4)                                                                                                                                    |                                                                                                                                                                                                                   |

Max 4 for similarity / difference with at least 1 similarity + 1 difference + 1mp for table

Reject: “Variety of cell types” → Fig2.2 one type; (a) two types / more than one type on basis that this is not a very significant feature

Reject: “Absence of cell wall” or “Absence of \_\_\_\_\_”

- (f) Calculate the actual diameter of the cell shown by the line **X** in Fig. 2.2.

Show all the steps in your calculation. [2]

1. measures line X correctly in mm or cm ; Reject m (MMO; ignore precision)
2. shows (their measurement divided by or  $\div$  or  $\times 700$ )  
AND  $\times 1000$  or  $10^3$  (mm) or 10000 or  $10^4$  (cm) or  $\times 10 \times 1000$  ;  
[Both  $\times 10$  AND  $\times 1000$  must be shown]

**X = 25mm**

**Actual diameter of cell shown by X =  $25/700 \times 1000 = 35.7 \mu\text{m} = 36 \mu\text{m}$**

.....  $\mu\text{m}$

[Total: 18]

- 3 The information on the label of a bottle of one brand of cleaning solution for contact lenses states that it contains enzymes which remove protein from contact lenses.

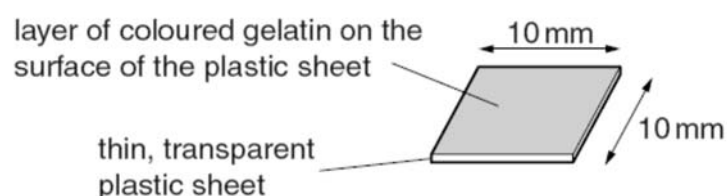
The contents of the solution are:

1. Subtilisin A, a protease with an optimum temperature of 60 °C and an optimum pH of 7.0–7.5.
2. Buffered saline, a solution with the same 'balance of ions' as body fluids.
3. Disodium EDTA, a preservative.

A student found a web site that gave the concentration of subtilisin A in different brands of contact lens cleaning solutions. These range between 20  $\mu\text{gcm}^{-3}$  and 100  $\mu\text{gcm}^{-3}$ .

The student simulated a dirty contact lens using the protein gelatin and a thin transparent plastic sheet. One side of the plastic sheet was dipped into melted gelatin containing a dye. The gelatin was then allowed to set.

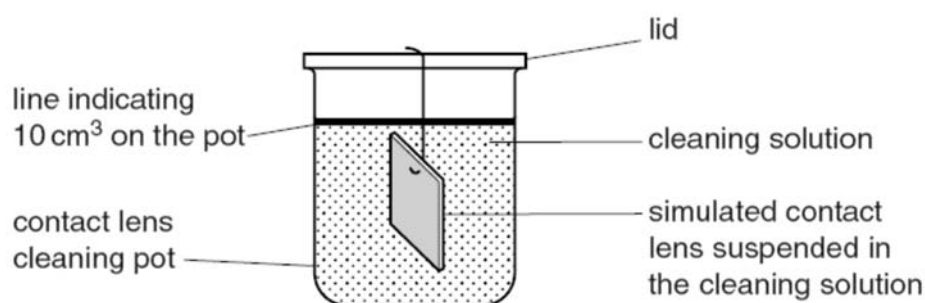
Fig. 3.1 shows a simulated contact lens.



**Fig. 3.1**

The student tested the activity of subtilisin A by recording the time taken for the coloured gelatin to be removed, leaving the transparent plastic sheet.

Fig. 3.2 shows the apparatus the student used.



**Fig. 3.2**



- (a) (i) On Fig. 3.3, add labels for the axes, including units, and show the expected shape of the curve for the factors investigated. [2]

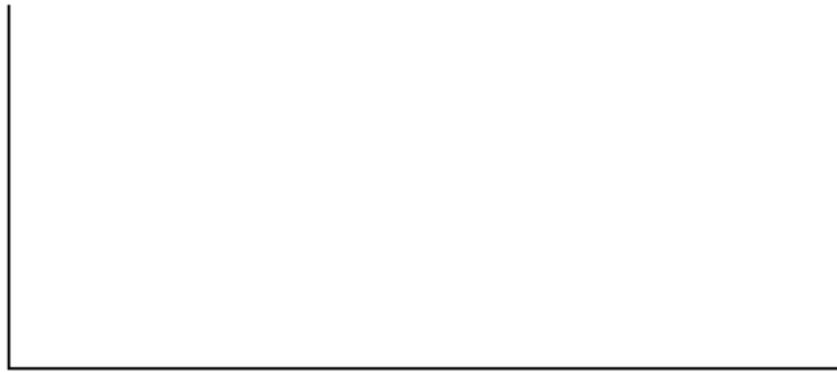
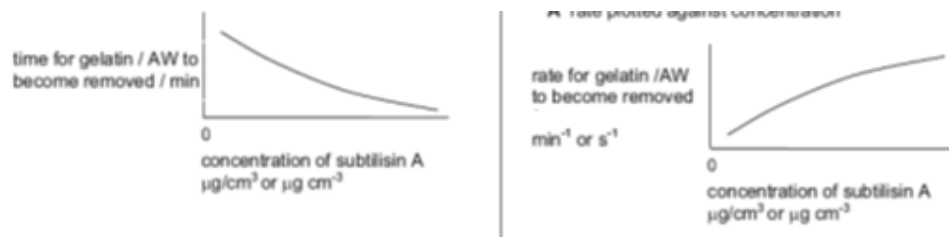


Fig. 3.3

1. axes correctly orientated with labels and units: x-axis, concentration of subtilisin A/ $\mu\text{gcm}^{-3}$  and y-axis, time for gelatin removal/s or min ; Reject: [ ] for concentration
2. line shows decrease as subtilisin A increases ;



- (ii) Describe how you would use the graph in (i) to find the actual concentration of subtilisin A in the contact lens cleaning solution. [1]

idea of: find the time for the gelatin to disappear (using the cleaning solution) on the y-axis and read the concentration from the x-axis ;

(b) In addition to your answer in (a), design an experiment to find the actual concentration of the protease present in the contact lens cleaning solution.

In your plan, you must use:

- 100  $\mu\text{gcm}^{-3}$  subtilisin A solution
- buffered saline solution
- coloured gelatin on 10 mm x 10 mm plastic sheet shown in Fig. 3.1
- stopwatch
- water bath
- cleaning pot shown in Fig. 3.2

You may select from the following apparatus and plan to use appropriate additional apparatus:

- normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, graduated pipettes, glass rods, etc.
- syringes
- filter funnels

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it
- be illustrated by relevant diagram(s), if necessary
- identify the independent and dependent variables
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible
- include layout of results tables with clear headings and labels
- use the correct technical and scientific terms
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[13]

**Mark Scheme**

*Award one mark for each of the following:*

1. state independent variable and uses 5 or more different concentrations ;
2. describe independent variable enzyme concentration must be between  $20 \mu\text{gcm}^{-3}$  and  $100 \mu\text{gcm}^{-3}$  ;
3. method:  
use **saline** (containing EDTA) to dilute enzyme stock solution in simple dilution/serial dilution ;
4. risk/safety ;
5. dependent variables:  
time taken for disappearance / breakdown / removal / AW, of, gelatin / protein / layer / colour (change) ;

*Award one mark for any of the following:*

6. independent variables:  
ref. to **correct volumes** of saline (containing EDTA) and of stock solution to give stated subtilisin concentration ;
7. controlled/fixed/standardized, variables to improve accuracy + reliability:  
identify two variables ;
8. +
9. controlled/fixed/standardized, variables to improve accuracy + reliability:  
**describe** how **two** identified variables are controlled ;;
10. control:  
refers to saline solution OR boiled/denatured enzymes OR (cleaning / saline) solution without, enzyme ;
11. method:  
incubate the, subtilisin / enzyme, solutions to, equilibrate / reach the test temperature (before adding the simulated contact lens) ;
12. method:  
use, stopwatch / timer, to record end point / AW ;
13. recording:  
results table for recording time taken for disappearance of colour and subtilisin A solution;
14. method:  
calculate rate as  $1/t$  ;
15. reliability:  
replicates and at least two repeats ;
16. accuracy:  
repeating with narrower range of concentration after identifying broad range within which optimum concentration lies/ more concentrations with smaller interval ;

[Total: 13]

## (a) Independent variables

**i. State and describe independent variable [1]**

- the independent variable is enzyme concentration

**ii. Describe independent variable [2]**

- 20, 40, 60, 80, 100  $\mu\text{gcm}^{-3}$  (equal interval) *R:concentration out of range*

## (b) Dependent variables

**State dependent variable [5]**

- the dependent variable is time taken for disappearance/breakdown of gelatin

## (c) Controlled/fixed/standardised, variables to improve accuracy or reliability

**i. Identifies at least two variables to be controlled [7]****ii. Describes how two identified variables are controlled [8,9]**

- total volume of enzyme and saline solution
  - kept constant at 10  $\text{cm}^3$  by topping up to the line on the tube
- temperature of reaction mixture
  - constant temperature for incubation, 60 °C
- area/size of (simulated) contact lens/plastic
  - kept constant at 10 mm x 10 mm
- thickness/mass/distribution/coverage of gelatin
  - kept constant at same thickness e.g. 1 mm OR
  - kept constant at same mass e.g. 1 g
- pH
  - use of, buffer saline, to keep pH constant / to control pH at pH 7.0–7.5

## (d) Scientific reasoning

**Describe the method with the scientific reasoning used to decide the method so that the results are as accurate and repeatable as possible**

- Time taken for subtilisin to digest/ protein gelatin is determined by the disappearance of coloured gelatin/decolourisation of gelatin
- Plot a graph of time taken for gelatin to decolourise OR rate of gelatin decolourisation against concentration of subtilisin
- use graph to read off the concentration of subtilisin A in contact lense solution

(e) Describe the experimental procedure (method)

*i. Use saline to dilute enzyme stock solution in simple/serial dilution [3]*

*ii. Correct volume of saline and stock solution [6]*

1. Prepare enzyme solutions of 5 different concentrations using simple dilution according to the table below. Label the containers accordingly.

| Concentration of subtilisin / $\mu\text{gcm}^{-3}$ | Volume of 100 $\mu\text{gcm}^{-3}$ subtilisin A solution / $\text{cm}^3$ | Volume of saline / $\text{cm}^3$ [3] |
|----------------------------------------------------|--------------------------------------------------------------------------|--------------------------------------|
| 20                                                 | 8.0                                                                      | 32.0                                 |
| 40                                                 | 16.0                                                                     | 24.0                                 |
| 60                                                 | 24.0                                                                     | 16.0                                 |
| 80                                                 | 32.0                                                                     | 8.0                                  |
| 100                                                | 40.0                                                                     | 0.0                                  |

2. Label 5 contact lens cleaning pots for each concentration.
3. Add 10  $\text{cm}^3$  of enzyme mixture into the cleaning pot labelled 20.

*iii. Plans suitable method to incubate the, subtilisin / enzyme, solutions to, equilibrate / reach the test temperature (before adding the simulated contact lens) [11]*

4. Leave this pot of enzyme mixture in a thermostatically controlled water bath to equilibrate at 60°C for 5 minutes
5. Place the lid with contact lens into the solution.

*iv. use, stopwatch / timer, to record end point [12]*

6. Start timing using a stopwatch.
7. + Record the time taken for gelatin film to completely decolourise. [12]
8. Repeat steps 3-7 to obtain another 2 readings. [15]
9. Repeat steps 3-8 for the other enzyme concentrations, including the unknown concentration of subtilisin A in contact lenses solution.
10. Repeat the entire experiment twice. [15]
11. Calculate the rate of reaction subtilisin A by using 1/t. [14]

(f) Control

*Refers to control experiment [10]*

- A control can be set up by replacing subtilisin A with 10  $\text{cm}^3$  of saline solution without enzyme OR boiled and cooled enzymes.

(g) Ensuring reliability and accuracy

*i. Describe how to ensure reliability (replicates and repeats) [15]*

(Reliability)

- Repeat two more times with fresh enzyme solution and substrate.

*ii. Describe how to ensure accuracy [16]*

(Accuracy)

- Repeating with more enzyme concentrations with smaller intervals.

## (h) Recording

***Shows how results are to be presented in the form of tables with independent (concentration) and dependent variables (time taken for decolourisation) in appropriate columns / rows [13]***

| Concentration of subtilisin/ $\mu\text{gcm}^{-3}$ | Time taken for disappearance of gelatin/s |           |           |         |
|---------------------------------------------------|-------------------------------------------|-----------|-----------|---------|
|                                                   | Reading 1                                 | Reading 2 | Reading 3 | Average |
|                                                   |                                           |           |           |         |
|                                                   |                                           |           |           |         |
|                                                   |                                           |           |           |         |
|                                                   |                                           |           |           |         |
|                                                   |                                           |           |           |         |

## (i) Risks/safety

***Refers to hazards and precaution [4]***

| Risks                                                                                 | Safety Precautions                                           |
|---------------------------------------------------------------------------------------|--------------------------------------------------------------|
| This is a <b>low risk investigation</b> and suitable safety precautions can be taken: |                                                              |
| Subtilisin could be a source of allergen/irritants to eyes or skin.                   | Wear goggles or gloves to prevent contact with eyes or skin. |
| EDTA could be a source of allergen/irritant to eyes or skin.                          | Wear goggles or gloves to prevent contact with eyes or skin. |
| Hot water could cause scalding.                                                       | Use a rag to hold beaker of hot water to prevent scalding.   |
| Glass wares are breakable and may cut.                                                | Keep glass wares away after use to prevent breakage.         |

[13]

[Total: 16]